

Relationship Between Renewable & Nonrenewable Energy with Consumption and Investment in South Asian Countries

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Introduction and Background

The energy and economic growth challenges of the 21st century for the emerging economies like the South Asian nations leave them in a bit of a dilemma how to ensure energy access to about one-third of the entire world population who are still deprived of uninterrupted energy connections with sustainable environmental policies in effect. While countries like India are rising as economic super giants and competing against already established massive economies from the West, they also have to secure their place into the global transition of a clean and carbon efficient system simultaneously. Energy pricing is also an issue in such countries since most developed countries levy taxes on electricity for environmental causes while their counterparts often subsidize this sector for social development. Renewable Energy can minimize greenhouse gas emissions, high energy prices, volatility of energy, dependence on energy import and it can improve energy security, provide socio-economic benefits while the decentralized nature of the renewable energy projects, according to Masud (2009) have the potential to provide electricity to remote and rural areas thereby help alleviating poverty and reducing the need to collect and burn biomass fuel.¹ Furthermore, renewable energy sources for the most part is cost effective and technologically apt but the energy produced through some of these sources are often required to be used with certain cautions or conditions. For instance, Bangladesh in recent years have revolutionized the solar energy production and consumption at the household level but the energy produced through the solar panels are more than often direct currents (DC) which are unable to support appliances that require constant electricity access. Another issue is that appliances that are supported by DC currents require certain connectivity, wire and board systems etc. to control for the voltages and heating issues to perform properly for their expected life periods, but because of using AC currents for so many years, the country has failed to produce those materials and more than often the DC connections are used through systems made for AC connections which cause problems to the devices used and the panels themselves. However, despite these issues, the country has gone on to set an unprecedented example of solar energy use and the other South Asian countries are also investing more readily into renewable sources than before in absolute terms mostly because of fossil fuels' high and fluctuating energy prices and solar, wind, and hydro energy's cost reductions. Solar energy, wind power, hydroelectricity, geothermal energy, biomass etc. are some of the mostly used renewable sources while fossil fuels such as coal, petroleum, natural gas etc. sources that have carbon as their main elements in them are considered as non-replenishable or non-renewable.

While the causality between energy use and economic growth is not a matter of a novel study, there have been broad disparities in the findings of the researches. Most of these works were also done on the aggregate energy consumption to GDP growth while only a handful of the studies recently like Apergis (2011)² and

Payne (2011), Salim et al. (2014)³, and Tugci et al. (2012)⁴ regarded energy consumption as renewable and nonrenewable separately before approaching their projects. Since the mentioned works were done on different regions of the world using different data sets and econometric approaches, the regional heterogeneity produced conflicting results. This paper not only disaggregates the energy and focuses on the renewable and nonrenewable energy share separately but also relates it to the consumption and investment growth than the entire GDP to exhibit separate impacts on growth rather than aggregating it. At a period when the South Asian countries are looking for ways to fill the energy gaps of their energy poor people while keeping the environmental qualities intact simultaneously, this research can do both contribute to the overall literature of sustainable development and put focus on coordinated and balanced growth of economy and renewable energy.

Literature Review

The Energy consumption and economic growth researches can be divided into four categories. First, the growth hypothesis states that an increase in energy consumption boosts up economic growth, and therefore, energy is an important input for output. The conservative hypothesis, secondly, views that there is a unidirectional causality from economic growth to energy consumption and hence changes in energy growth doesn't affect economic growth. Next, the feedback hypothesis considers that there is a bidirectional causality between the discussed variables while finally the neutrality hypothesis assumes that there exists no causality between them. Growth hypothesis has been confirmed by Mbarek et al. (2017)⁵, Bowden and Payne (2009)⁶, Narayan and Smith (2008)⁷, Tang et al. (2016)⁸ etc. for countries such as USA, OECD, non-OECD, G7, South American Countries, Vietnam etc. for different periods. Conservative hypothesis was proved to be valid by Lise and Montford (2007)⁹ for Turkey (1970-2003) and Huang et al. (2008)¹⁰ for 82 countries (1982-2002). Feedback hypothesis was validated by Saidi et al. (2017)¹¹, for 53 nations, Fuinhas and Marques (2012)¹² for 5 European countries, Eggoh et al. (2011)¹³ for 21 African countries etc. Co-existence of combined hypotheses were also found in several researches. For instance, Soytas and Sari (2006)¹⁴ found evidence of growth, conservative and feedback hypotheses in their research on G7 countries and 10 emerging markets while Lee (2006)¹⁵ and Akinio (2008)¹⁶ found the evidence of conservative, feedback and neutrality hypotheses in their respective studies on 11 major industrial and Sub-Saharan African nations.

For only renewable energy and GDP, growth hypothesis was ascertained by Tiwari (2011)¹⁷, Fang (2011)¹⁸, Payne (2011)¹⁹, and Chien and Hu (2007)²⁰ for India, China, USA, and 45 countries respectively. Sadorsky (2009)²¹ revealed the existence of conservative hypothesis for 18 emerging countries while Chen (2018)²² and Yang et al (2016)²³ also found evidence that it is growth that leads to increase in renewable energy and not otherwise. The feedback hypothesis, on the other hand, was proved by Apergis (2010)²⁴ and Payne

(2010)²⁵ for OECD and Eurasian countries etc. Rahman and Velayutham (2019)²⁶ most recently concluded that conservative hypothesis is indeed valid for the South Asian countries. In another interesting study, Nguyen and Kakinada (2019)²⁷ argued that there exists a negative relationship between renewable energy consumption and output in low-income countries while the opposite appears to be true for the high-income ones.

For South Asian countries, Akhmats and Zaman (2013)²⁸ conducted a study over the data from 1975-2010 and found that nuclear energy Granger causes economic growth in Nepal and Pakistan while commercial energy consumption Granger causes economic growth in Bangladesh, Bhutan, Maldives, Nepal and Sri Lanka; gas consumption Granger causes economic growth in Bangladesh, Bhutan, India and Maldives; electricity consumption Granger causes economic growth in India and Sri Lanka, and finally, coal consumption Granger causes economic growth in Bangladesh, Bhutan, Nepal and Sri Lanka.

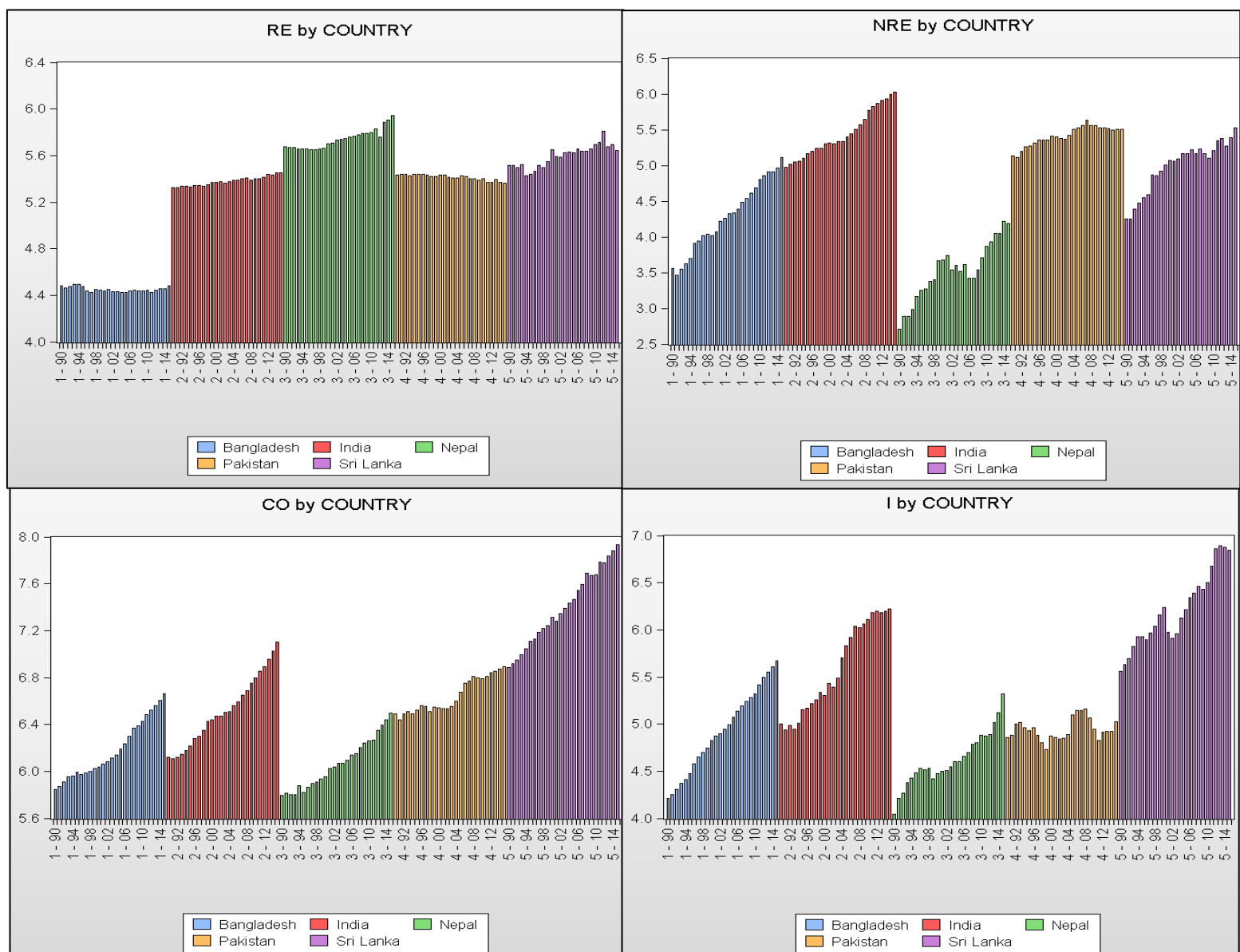
As mentioned earlier, despite a lot of work in this sector of study, there has been no work that tried to find a causal relationship between disaggregated energy use with consumption and investment in a broad manner and hence this research aspires to fill that significant gap. Since the majority of the solar energy which are one of the greatest contributors to renewable energy sources in this region are produced and consumed at the household level, it's important to find the impact on consumer spending due to renewable energy use separately and to see if it affects the investors as much, given that many low income small households and small startup investors in rural areas are now demanding energy more than ever with very little access to existing energy sources.

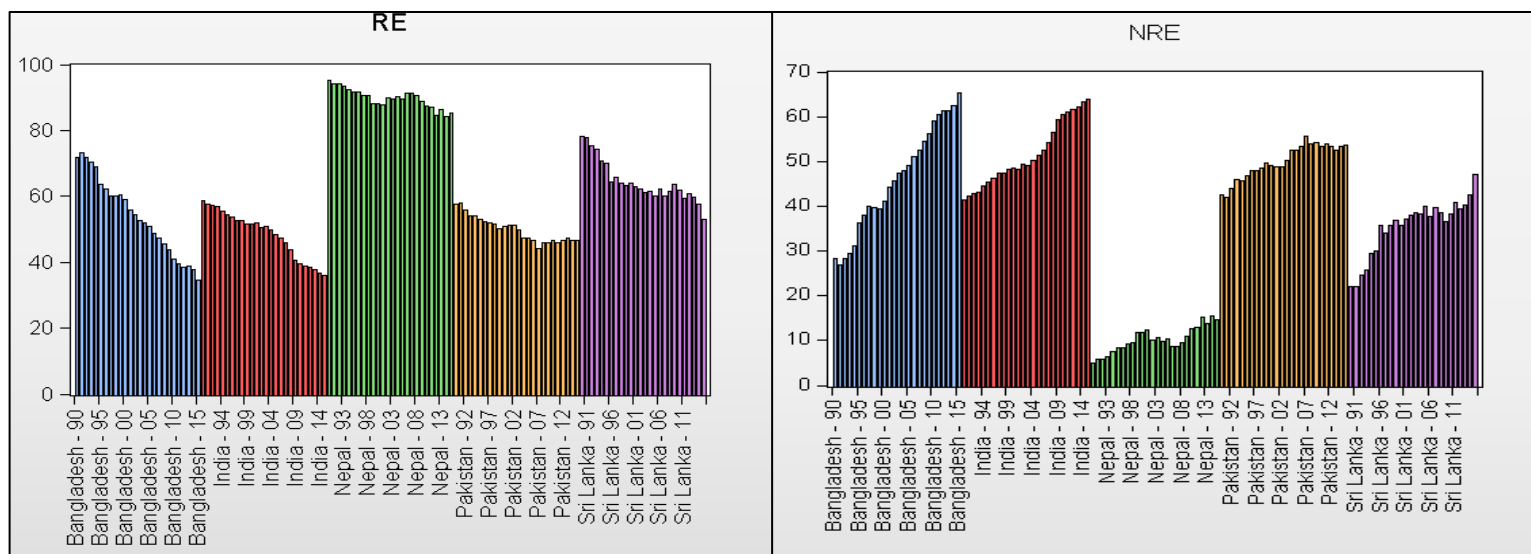
Data and Descriptive Statistics

All the quantitative data for the study were collected from the World Bank database. Data ranging from 1990 to 2015 were taken for five South Asian countries alphabetically Bangladesh, India, Nepal, Pakistan, and Sri Lanka from the World Development Indicators (WDI) for panel estimations.²⁹ Level data for renewable and nonrenewable energy aren't readily available and hence renewable energy consumption (REC) as a percentage of total energy consumption was taken from the source and nonrenewable energy consumption (NREC) was established by deducting REC from 100%. In addition, total per capita energy consumption in KG of oil equivalent was taken which was then multiplied with the previous percentage variable REC and NREC to find the per capita renewable and nonrenewable energy consumption in level units for the concerned countries. These modified variables are used as the primary exploratory variables in this study. Among the two resulting variables for two separate models, the first one is per capita consumption for which data for Gross Domestic Product in constant 2010 USD were taken as well as final consumption as a percentage of GDP was collected simultaneously and then the two were multiplied. The

result of the products was then divided by the total population to find the level data for per capita final consumption. For the other dependent variable per capita investment, an identical approach was followed while gross fixed capital formation was approximated as an indicator for investment. Also, since income data are not easily accessible, expenditure data were taken instead as their replacements.

Regarding control variables financial development or broad money is considered to be one which was acquired by the domestic credit to private sector as a percentage of GDP. Trade openness ratio which is the sum of export and import as a ratio of GDP is also used as another control variable in the research. Import and Export of goods and services were taken in constant 2010 USD and they were added up and divided by total GDP to achieve the ratio.





The data from the first four figures (log applied models) indicates that Bangladesh has the least renewable energy use per capita compared to its other four neighboring countries while Nepal has the highest and Nepal also has the lowest use of per capita nonrenewable energy use while India has the highest. Sri Lanka, on the other hand, has the highest consumption and investment per capita among the considered countries while Nepal has the lowest on average. From the last two figures where it shows the relative share of total renewable and nonrenewable energy as a percentage of total energy share for the countries show that as shares it's the nonrenewable energy that is increasing rather than the earlier although in absolute terms, investments in renewable sources have increased by quite some margin over the years.

Model and Methodology

Since the objective of the study is to investigate and expose the relationship between consumption and Investment growth to renewable and nonrenewable energy while controlling for a couple of important variables, the following log linear models are formed to test and capture the potential short run and long run relationships between the dependent and independent variables.

$$\ln C_{it} = \alpha_i + \beta_{1i} \ln RE_{it} + \beta_{2i} \ln NRE_{it} + \beta_{3i} \ln FD_{it} + \beta_{4i} \ln TO_{it} + \xi_{it} \text{-----(I)}$$

$$\ln I_{it} = \alpha_i + \beta_{1i} \ln RE_{it} + \beta_{2i} \ln NRE_{it} + \beta_{3i} \ln FD_{it} + \beta_{4i} \ln TO_{it} + \xi_{it} \text{-----(II)}$$

In models (I) and (II), subscript i denotes the countries, t denotes the years. C stands for the per capita consumption, I stands for per capita investment, the coefficients $\beta_1, \beta_2, \beta_3, \beta_4$, represent magnitudes of their

respective variables, while the independent variables RE and NRE and control variables FD and TO symbolize per capita renewable and nonrenewable energy use, financial development and trade openness respectively. Here, expressing the target and principal exploratory variables in per capita form and applying log to the models extenuates heteroskedasticity. The use of panel data naturally reduces multicollinearity among the exploratory variables and increases the degrees of freedom which drives to better estimations. It also controls the individual heterogeneity and identifies the effects that couldn't be detected through time series or cross-sectional analysis.³⁰

For this particular investigation, panel unit root tests have been performed first with the alternative hypothesis that there is no unit root in the series. A total of 5 panel unit root tests have been executed to enhance the robustness of the results. Among them, Levin, Lin, Chu (LLC) and Breitung tests assume common unit root process while Im, Pesaran, Shin (IPS), Fisher ADF and Fisher PP tests assume individual unit root process with null hypothesis that there is unit root or non-stationarity among the variables. A couple of panel cointegration tests followed then to find potential long run relationships among the variables. Pedroni³¹ and Kao³² are two Engle-Granger multi steps residual-based panel cointegration³³ tests with null hypothesis of no cointegration which were performed to check for long run cointegration among the variables and check the robustness of the estimated results.³⁴

After that, Fully Modified Ordinary Least Squares (FMOLS) and Dynamic Ordinary Least Squares (DOLS) were used to establish the long run equilibrium association among the variables. A number of research^{35,36,37} suggests that the FMOLS method is the most appropriate approach to be applied in the presence of cointegration of panel data analysis which corrects serial correlation, endogeneity, and simultaneity bias.²⁶ DOLS was also adopted to further the robustness of the findings as many econometricians including Kao and Chiang (2000)³⁸ recommend DOLS over FMOLS as a method in small samples using Monte Carlo simulations which has better sample properties than the panel FMOLS estimators.

Next, Vector Auto Regressive Model was run to detect the optimal lag length and using that, Vector Error Correlation Method (VECM) was applied to test the existence of causal links among the variables as it has been used in many studies concerning growth and energy consumption recently including Fan & Hao (2019)³⁹, Mahadevan & Asafu-Adjaye (2007)⁴⁰, Hao et al (2018)⁴¹ etc. Then the ECM system was run as a regression to find the probability readings for the VECM coefficients to find the significance and therefore the relevance of the results. Next, dynamic impact or impulse response function analysis was performed among the variables to shed lights on the findings of the VECM results. It expresses the dynamic relationship among the variables, describes the impact of one standard deviation shock in one variable onto another variable considering the entire system while adjusting the lag periods, stabilizing the process and providing the corresponding short and long run directions.³⁹

Finally, Dumitrescu-Hurlin causality test was approached to find the direction of causality among the variables.⁴² Since FMOLS and DOLS don't provide information regarding causal relationships among variables and VECM doesn't imply directions of causality, this study adopted a relatively recently introduced Dumitrescu-Hurlin causality test, which is essentially a type of Granger causality test, to establish directional causal relationship in between the variables. This causality test has an upper hand over the traditional Granger causality test as it not only considers the heterogeneity of the regression model which is used to test the Granger causality but also the heterogeneity of the causal relationships.⁴³ It also tests for individual coefficients rather than the common ones while testing for the causality.

Empirical Tests and Analysis

Panel Unit Root Test

	Tests	Ln C pc	Ln I pc	Ln RE pc	Ln NRE pc	Ln FD	Ln TO
	LLC	0.27	0.60	-1.89**	0.50	0.44	1.74
	Breitung	2.97	-0.38	0.32	0.55	-0.16	2.66
Level	IPS	1.40	-0.33	-1.16	0.07	-0.61	2.66
	ADF- Fisher	8.89	12.01	15.69	10.14	13.22	3.48
	PP- Fisher	9.42	6.14	14.66	8.40	12.36	3.34
	LLC	-8.04***	-5.97***	-9.25***	-5.68***	-5.38***	-8.19***
	Breitung	-3.77***	-1.98**	-5.70***	-4.63***	-3.69***	-5.17***
First Difference	IPS	-7.79***	-5.07***	-10.36***	-6.19***	-5.02***	-6.45***
	ADF- Fisher	65.59***	40.38***	87.78***	50.33***	43.86***	51.07***
	PP- Fisher	446.63***	32.96***	349.50***	60.52***	57.01***	57.71***

Here the tests have been conducted with individual intercepts and trend included to meet all the tests' criteria and maximum lag length was selected through Schwarz Information Criterion (SIC). All the variables are found to be non-stationary at level but they all reject the null hypothesis at the first difference or I (1). With all the variables being stationary at I (1), they could be prepared for long run cointegration and causality tests which otherwise wouldn't be economically sensible.

Johansen Panel Cointegration Test

Model (I)

Pedroni Residual Cointegration Test	Statistics
<i>Within Dimensions</i>	
Panel v-Statistics	0.64
Panel rho-Statistics	-0.67
Panel PP-Statistics	-2.83***
Panel ADF-Statistics	-2.84***
<i>Between Dimensions</i>	
Group rho-Statistics	-0.11
Group PP-Statistics	-3.37***
Group ADF-Statistics	-3.33***
Kao Residual Cointegration Test	
ADF	-1.79**

Model (II)

Pedroni Residual Cointegration Test	Statistics
<i>Within Dimensions</i>	
Panel v-Statistics	1.29*
Panel rho-Statistics	-0.19
Panel PP-Statistics	-1.57*
Panel ADF-Statistics	-1.37*
<i>Between Dimensions</i>	
Group rho-Statistics	0.55
Group PP-Statistics	-1.66**
Group ADF-Statistics	-3.45***
Kao Residual Cointegration Test	
ADF	-2.07**

Individual intercept was used for deterministic trend specification and again SIC was used with maximum lag selected automatically. Long run cointegration was found in both the models for 5% significance level in Kao test while model (I) proved to be long-term cointegrated at 1% level according to Pedroni test in 4 criteria and model (II) was found to be long-term cointegrated for a mixed significance level in a total of 5 criteria implying that consumption has slightly better long run correlation with the regressive variables than investment.

FMOLS, DOLS

Model (I)

Variables	FMOLS Coefficients	DOLS Coefficients
RE	0.661435***	0.644702***
NRE	0.606123***	0.601880***
FD	-0.027029	0.005650
TO	-0.114955**	-0.063115

Model (II)

Variables	FMOLS Coefficients	DOLS Coefficients
RE	0.342648***	0.314034***
NRE	0.659959***	0.761813***
FD	0.078855	-0.014495
TO	-0.121029	-0.143149

The long run covariance estimates are performed using Bartlett Kernel and Newey-West fixed bandwidth for individual coefficient covariances based on SIC automatic leads and maximum lag specifications. From the results of both the approaches, having all other variables fixed, it is observed that both renewable and nonrenewable energy have positive effects on consumption and investment in the long run at 1% significance level while renewable energy influences consumption growth more comparatively as nonrenewable energy does for investment according to the coefficients estimates. More specifically according to FMOLS, a 1% increase in renewable and nonrenewable energy keeping all other variables constant would increase consumption by 0.66% and 0.61% respectively while for investment the increase is 0.34% and 0.66% respectively. However, these tests do not stipulate causation or direction of causal relationships among variables, but only confirm long term relationship among them at a certain confidence level.

Vector Auto Regressive Model

VAR Lag Order Selection Criteria Endogenous variables: CO RE NRE FD TO Exogenous variables: C Date: 12/30/19 Time: 22:21 Sample: 1990 2015 Included observations: 90							VAR Lag Order Selection Criteria Endogenous variables: I RE NRE FD TO Exogenous variables: C Date: 12/30/19 Time: 21:27 Sample: 1990 2015 Included observations: 90						
Lag	LogL	LR	FPE	AIC	SC	HQ	Lag	LogL	LR	FPE	AIC	SC	HQ
0	-212.7234	NA	8.69e-05	4.838298	4.977177	4.894302	0	-238.4100	NA	0.000154	5.409111	5.547990	5.465115
1	702.5793	1708.565	2.22e-13*	-14.94621*	-14.11294*	-14.61018*	1	618.2850	1599.164	1.45e-12*	-13.07300*	-12.23973*	-12.73698*

Vector Auto Regressive Model (VAR) was run for both models with all the variables set as endogenous variables and then lag structure and subsequently lag length criteria was checked to find the optimal lag length for the SIC. For both models, optimal lag was found to be 1 for SIC.

Vector Error Correction Method

According to VECM of model (I), the error correction term $ECT_{t-1} = \ln CO_{t-1} - 0.576051 \ln RE_{t-1} + 0.495342 \ln NRE_{t-1} + 5.049812 \ln FD_{t-1} - 0.186287 \ln TO_{t-1} - 22.78718$.

For model (II), $ECT_{t-1} = \ln I_{t-1} - 1.041145 \ln RE_{t-1} - 1.764436 \ln NRE_{t-1} - 5.451258 \ln FD_{t-1} + 9.315189 \ln TO_{t-1} + 36.54974$.

Model (I)

Cointegrating Eq:	CointEq1
CO(-1)	1.000000
RE(-1)	-0.576051 (1.05566) [-0.54568]
NRE(-1)	0.495342 (0.50622) [0.97851]
FD(-1)	5.049812 (1.15080) [4.38809]
TO(-1)	-0.186287 (1.08901) [-0.17106]
C	-22.78718

Model (II)

Cointegrating Eq:	CointEq1
I(-1)	1.000000
RE(-1)	-1.041145 (2.22686) [-0.46754]
NRE(-1)	-1.764436 (1.06899) [-1.65056]
FD(-1)	-5.451258 (2.40859) [-2.26325]
TO(-1)	9.315189 (2.28262) [4.08091]
C	36.54974

Error Correction:	D(CO)	D(RE)	D(NRE)	D(FD)	D(TO)	Error Correction:	D(I)	D(RE)	D(NRE)	D(FD)	D(TO)
CointEq1	0.004490 (0.00120) [3.75377]	0.002213 (0.00123) [1.79784]	-0.001996 (0.00302) [-0.66148]	-0.013315 (0.00541) [-2.46069]	-0.000115 (0.00436) [-0.02636]	CointEq1	-0.001355 (0.00150) [-0.90309]	-0.000109 (0.00064) [-0.16879]	0.001541 (0.00152) [1.01338]	0.006153 (0.00267) [2.30252]	-0.008016 (0.00207) [-3.87469]
D(CO(-1))	-0.102247 (0.09349) [-1.09366]	0.150674 (0.09620) [1.56632]	-0.109617 (0.23586) [-0.46475]	0.085352 (0.42295) [0.20180]	-0.409008 (0.34077) [-1.20024]	D(I(-1))	0.298097 (0.09714) [3.06859]	0.008463 (0.04170) [0.20294]	0.051901 (0.09848) [0.52702]	0.381775 (0.17300) [2.20678]	0.266857 (0.13394) [1.99235]
D(RE(-1))	-0.043259 (0.08702) [-0.49714]	-0.324666 (0.08953) [-3.62615]	-0.001894 (0.21953) [-0.00863]	-0.671561 (0.39366) [-1.70594]	0.048195 (0.31717) [0.15195]	D(RE(-1))	-0.284085 (0.22202) [-1.27953]	-0.292422 (0.09531) [-3.06819]	-0.070465 (0.22507) [-0.31308]	-1.105629 (0.39539) [-2.79630]	-0.142093 (0.30612) [-0.46418]
D(NRE(-1))	0.037207 (0.03747) [0.99303]	0.053501 (0.03855) [1.38773]	-0.073798 (0.09453) [-0.78071]	0.180968 (0.16951) [1.06762]	0.139014 (0.13657) [1.01789]	D(NRE(-1))	0.225142 (0.09350) [2.40790]	0.058222 (0.04014) [1.45056]	-0.090917 (0.09479) [-0.95918]	0.136454 (0.16651) [0.81948]	0.196315 (0.12892) [1.52280]
D(FD(-1))	-0.013625 (0.01775) [-0.76745]	0.004475 (0.01827) [0.24496]	0.136586 (0.04479) [3.04953]	0.092345 (0.08032) [1.14976]	0.007550 (0.06471) [0.11666]	D(FD(-1))	-0.020222 (0.04334) [-0.46653]	0.014195 (0.01861) [0.76292]	0.127891 (0.04394) [2.91061]	0.054622 (0.07719) [0.70763]	-0.014925 (0.05976) [-0.24974]
D(TO(-1))	0.004186 (0.02598) [0.16113]	-0.021494 (0.02673) [-0.80408]	0.025376 (0.06554) [0.38717]	0.042187 (0.11753) [0.35894]	0.189328 (0.09470) [1.99933]	D(TO(-1))	0.009071 (0.06788) [0.13364]	-0.015740 (0.02914) [-0.54016]	0.019623 (0.06881) [0.28516]	-0.003038 (0.12088) [-0.02513]	0.069277 (0.09359) [0.74020]
C	0.034675 (0.00395) [8.78720]	-0.001636 (0.00406) [-0.40292]	0.049148 (0.00996) [4.93687]	0.026064 (0.01785) [1.46001]	0.022729 (0.01438) [1.58021]	C	0.021712 (0.00815) [2.66414]	0.001905 (0.00350) [0.54455]	0.044996 (0.00826) [5.44628]	0.018287 (0.01451) [1.26001]	0.000199 (0.01124) [0.01768]

The above equations formed from the tables from the first panel signify long run causal relations among the cointegrated variables. The relations are in elasticity terms as log is applied to the model. The tables in the second panel signify the short run relationships and long run adjustment speeds. It is observed from the tables that renewable energy is negatively associated with consumption and investment in the short run while a positive relation exists between them in the long run. On the other hand, despite having positive affiliation with both in the short run, nonrenewable energy has negative and positive correlations with consumption and investment in the long run respectively. Perhaps it's because in the longer period, consumption of nonrenewable energy causes too much pollution which hampers consumption growth while it doesn't affect the investors as much.

Model (I)

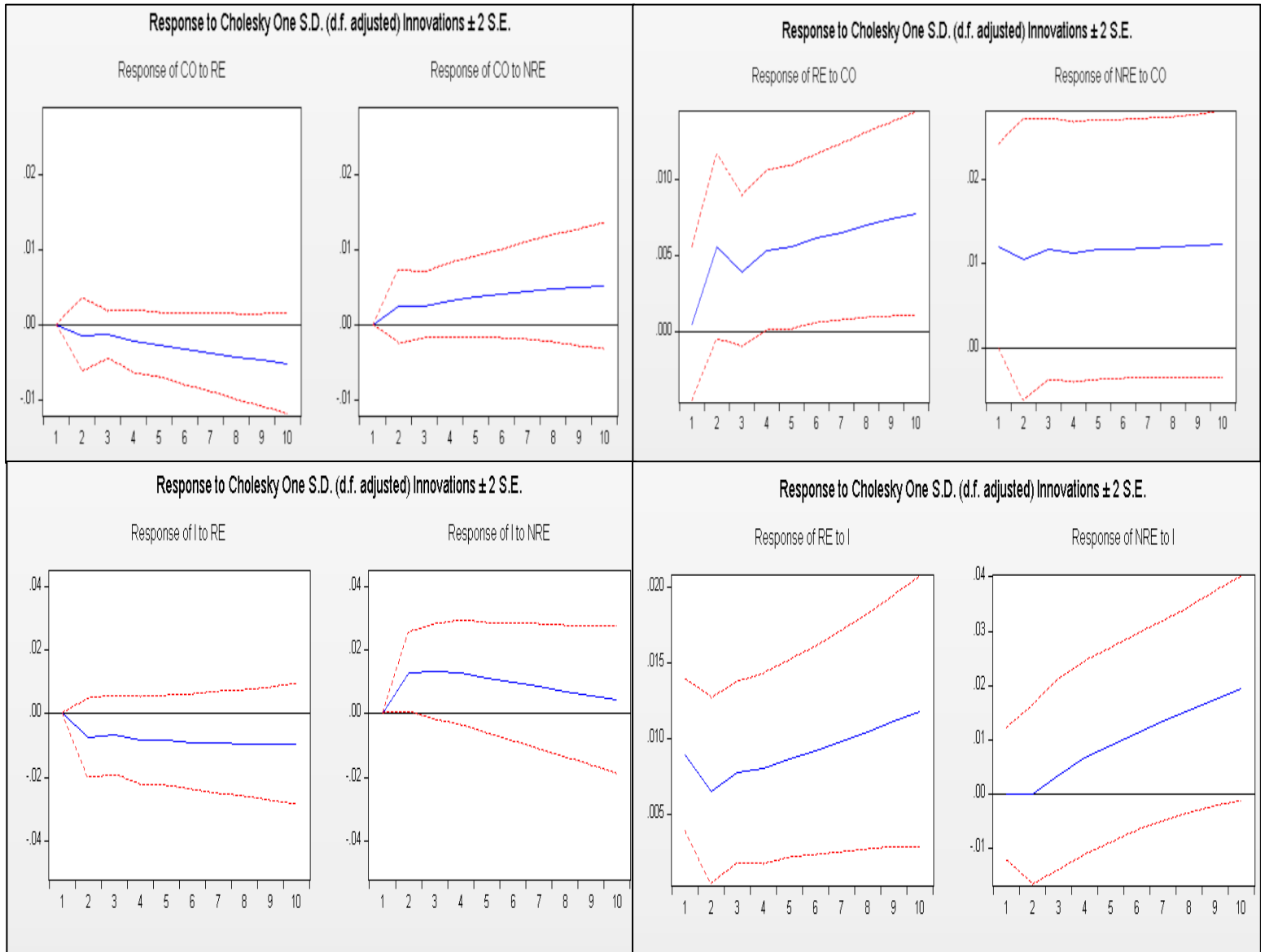
Equation: $D(CO) = C(1) * (CO(-1) - 0.576051109969 * RE(-1) + 0.495341559779 * NRE(-1) + 5.04981156624 * FD(-1) - 0.186287211332 * TO(-1) - 22.7871808252) + C(2) * D(CO(-1)) + C(3) * D(RE(-1)) + C(4) * D(NRE(-1)) + C(5) * D(FD(-1)) + C(6) * D(TO(-1)) + C(7)$				
Estimation Method: Least Squares Date: 12/29/19 Time: 19:23 Sample: 1992 2015 Included observations: 120 Total system (balanced) observations 600				
	Coefficient	Std. Error	t-Statistic	Prob.
C(1)	0.004490	0.001196	3.753772	0.0002
C(2)	-0.102247	0.093491	-1.093658	0.2746
C(3)	-0.043259	0.087017	-0.497141	0.6193
C(4)	0.037207	0.037468	0.993027	0.3211
C(5)	-0.013625	0.017754	-0.767448	0.4431
C(6)	0.004186	0.025980	0.161128	0.8721
C(7)	0.034675	0.003946	8.787196	0.0000

Model (II)

Equation: $D(I) = C(1) * (I(-1) - 1.04114539212 * RE(-1) - 1.76443624506 * NRE(-1) - 5.45125750225 * FD(-1) + 9.31518856515 * TO(-1) + 36.5497405343) + C(2) * D(I(-1)) + C(3) * D(RE(-1)) + C(4) * D(NRE(-1)) + C(5) * D(FD(-1)) + C(6) * D(TO(-1)) + C(7)$				
Estimation Method: Least Squares Date: 12/29/19 Time: 19:28 Sample: 1992 2015 Included observations: 120 Total system (balanced) observations 600				
	Coefficient	Std. Error	t-Statistic	Prob.
C(1)	-0.001355	0.001500	-0.903090	0.3669
C(2)	0.298097	0.097145	3.068586	0.0023
C(3)	-0.284085	0.222022	-1.279534	0.2012
C(4)	0.225142	0.093502	2.407898	0.0164
C(5)	-0.020222	0.043344	-0.466534	0.6410
C(6)	0.009071	0.067880	0.133638	0.8937
C(7)	0.021712	0.008150	2.664138	0.0079

The probability estimates for the coefficients give relevance to the values and from the VECM, it's found that consumption has long run causal relationship with renewable and nonrenewable energy at 1% significance level while investment has short run causal relationship with nonrenewable energy at 5% level only. In other words, there exists a positive long run causal relationship between consumption and renewable energy and a negative one between consumption and nonrenewable energy while investment and nonrenewable energy have a positive causal relationship between them in the short run. All the other causal relationships are found to be insignificant according to VECM. Perhaps it's because investment theoretically is a more unstable variable because of its sensitivity towards interest rates and other dynamic variables, it doesn't achieve long run convergence and hence doesn't have long run adjustments in the model at a significant level.

Impulse Response Function Analysis



The dynamic impact or impulse response function analysis was carried on considering all the variables but only the relevant ones are shown and discussed here. According to the graphs, the impact of renewable energy on consumption is negative in both short run and long run although the magnitude is relatively small while nonrenewable energy has an insignificant positive effect on it. On the other hand, consumption positively and sharply prompts use of renewable energy while its impact on nonrenewable energy is stale in the long run. This relationship from renewable energy to consumption is consistent with the VECM findings while the earlier relationship the other way around isn't backed by it. Hence, the causality seems to be from renewable energy to consumption instead of the opposite way emphasizing the potential

existence of the conservative hypothesis. Similarly, for the second model, the impact of energy on investment is not found to be consistent with the significant estimates of VECM, rather the undisputed positive relation from nonrenewable energy to investment seems causally reasonable and consistent again reemphasizing conservative hypothesis. The impulse response, along with VECM, promptly implies that it's consumption that causes renewable energy to rise while investment causes nonrenewable energy to go up. It makes proper economic sense as with more investment, there would be more demand for energy and since nonrenewable energy is more traditional and require less installation cost, more investment is supposed to promote more nonrenewable energy extraction and use. But since renewable energy has long term economic and environmental advantage over nonrenewable resources, consumers demand the earlier more than the latter. On the other hand, reducing the use of conventional energy sources and substituting them with the newly renewable ones are bound to slow economic and therefore consumption and investment growth in the short run which can explain the negative impact of renewable energy on consumption and investment in the first half.³⁹

Dumitrescu-Hurlin Panel Causality Test

Model (I)

Model (II)

Null Hypothesis:	W-Stat.	Zbar-Stat.	Prob.	Null Hypothesis:	W-Stat.	Zbar-Stat.	Prob.
RE does not homogeneously cause CO	0.29...	-1.06691	0.2860	RE does not homogeneously cause I	2.06...	1.27891	0.2009
CO does not homogeneously cause RE	11.0...	13.2854	0.0000	I does not homogeneously cause RE	1.73...	0.85100	0.3948
NRE does not homogeneously cause CO	2.29...	1.58538	0.1129	NRE does not homogeneously cause I	0.85...	-0.33116	0.7405
CO does not homogeneously cause NRE	1.41...	0.42159	0.6733	I does not homogeneously cause NRE	4.62...	4.69499	3.E-06

The test statistics along with the probability values of the causality tests are calculated using Monte-Carlo simulations following Kocak & Sarkgunesi (2017)⁴⁴. Here also, the variables of interest are shown only. The results indicate that there exists a positive unidirectional Granger causality running from consumption to renewable energy and investment to nonrenewable energy at 1% and 5% significance levels respectively. These outcomes are completely consistent with the ones of FMOLS, DOLS, VECM and dynamic impact analysis even with identical significance level concluding that conservative hypothesis indeed holds in the case of South Asian countries.

Concluding Remarks and Policy Implications

This paper was dedicated to explore the relationship between renewable and nonrenewable energy with consumption and investment growth in the South Asian region. A dataset ranging from 1990 to 2015 was used for 5 SAARC countries (130 balanced sample data) and a range of descriptive and panel econometric tests were conveyed to reach the conclusion that consumption and investment positively affects renewable and nonrenewable energy growth respectively supporting the previously hypothesized conservative hypothesis for this area which states that economic growth boosts energy growth. The main contribution of this research to the previously published papers is double-fold: To confirm the presence of conservative assumption for this territory among the four possible options and also to pinpoint which factor of economic growth accelerates which category of energy, something that hasn't been done before in the best of my knowledge.

Since the fundamentals of conservative hypothesis is that change in energy production or consumption doesn't affect economic growth, so it makes all this essential for these countries to promote renewable energy as a substitute of nonrenewable energy as fast as possible since it doesn't endanger their existing economic prosperity. It is not possible to address climate change issues or ensure sustainable economic growth and persistent energy security without switching from nonrenewable energy sources to the renewable ones and the sooner the better it is for the entire world. But not all the countries have the privilege of making that choice because of the severe economic consequences they'll have to bear once they make that shift. Since the countries in question do have that choice, their governments should provide incentives like investment subsidies, tax cuts or other tax benefits, installation discounts to produce renewable energy. Proper initiatives should also be taken through public and private partnerships to ease up and develop the market accessibility for renewable energy. Public awareness should be raised as well by both governments and NGOs so that people reduce their dependency upon the traditional energy sources and open up towards using renewable sources more while businesses should be encouraged and regulated to produce materials for the systems to access renewable sources smoothly. Only then, a clean energy revolution could be translated into reality and the energy sector would be modernized in its true sense.

References

- ¹ Masud J. Clean energy development in Pakistan. ADB climate change and energy workshop. Bangkok: Hagler Bailly Pakistan; 2009.
- ² N. Apergis, J.E. Payne, Renewable and non-renewable electricity consumption growth nexus: evidence from emerging market economies, *Appl. Energy* 88 (12) (2011) 5226 - 5230.
- ³ R.A. Salim, K. Hassan, S. Shafiei, Renewable and non-renewable energy consumption and economic activities: further evidence from OECD countries, *Energy Econ.* 44 (2014) 350 - 360.
- ⁴ C.T. Tugcu, I. Ozturk, A. Aslan, Renewable and non-renewable energy consumption and economic growth relationship revisited: evidence from G7 countries, *Energy Econ.* 34 (2012) 1942 - 1950.
- ⁵ M.B. Mbarek, K. Saidi, M.M. Rahman, Renewable and non-renewable energy consumption, environmental degradation and economic growth in Tunisia, *Qual. Quantity* 52 (3) (2017) 1105 - 1119.
- ⁶ N. Bowden, J.E. Payne, The causal relationship between U.S. energy consumption and real output: a disaggregated analysis, *J. Policy Model.* 31 (2) (2009) 180 - 188.
- ⁷ P.K. Narayan, R. Smyth, Energy consumption and real GDP in G7 countries: new evidence from panel cointegration with structural breaks, *Energy Econ.* 30 (5) (2008) 2331 - 2341.
- ⁸ C.F. Tang, B.W. Tan, I. Ozturk, Energy consumption and economic growth in Vietnam, *Renew. Sustain. Energy Rev.* 54 (2016) 1506 - 1514.
- ⁹ W. Lise, K. Montfort, Energy consumption and GDP in Turkey: is there a cointegration relationship? *Energy Econ.* 29 (6) (2007) 1166 - 1178.
- ¹⁰ B.N. Huang, M.J. Hwang, C.W. Yang, The causal relationship between energy consumption and GDP growth revisited: a dynamic panel data approach, *Ecol. Econ.* 67 (1) (2008) 41 - 54.
- ¹¹ K. Saidi, M.M. Rahman, M. Amamri, the causal nexus between economic growth and energy consumption: new evidence from a global panel of 53 countries, *Sustain. Cities Soc.* 33 (2017) 45 - 56.
- ¹² J.A. Fuinhas, A.C. Marques, Energy consumption and economic growth nexus in Portugal, Italy, Greece, Spain and Turkey: an ardl bounds test approach (1965e2009), *Energy Econ.* 34 (2) (2012) 511 - 517.
- ¹³ J.C. Eggoh, C. Bangake, R. Christophe, Energy consumption and economic growth revisited in African Countries, *Energy Policy* 39 (11) (2011) 7408 - 7421.
- ¹⁴ U. Soytaş, R. Sari, Energy consumption and income in G-7 countries, *J. Policy Model.* 28 (7) (2006) 739 - 750.
- ¹⁵ C.C. Lee, the causality relationship between energy consumption and GDP in G-11 countries revisited, *Energy Policy* 34 (9) (2006) 1086 - 1093.
- ¹⁶ A.E. Akinlo, Energy consumption and economic growth: evidence from 11 Sub-Sahara African countries, *Energy Econ.* 30 (5) (2008) 2391 - 2400.
- ¹⁷ A.K. Tiwari, A structural VAR analysis of renewable energy consumption, real GDP and CO2 emissions: evidence from India, *Econ. Bull.* 31 (2) (2011) 1793 - 1806.
- ¹⁸ Y. Fang, Economic welfare impacts from renewable energy consumption: The China experience, *Renew. Sustain. Energy Rev.* 15 (2011) 5120 - 5128.
- ¹⁹ J.E. Payne, On biomass energy consumption and real output in the US, *Energy Sources Part B* 6 (1) (2011) 47 - 52.
- ²⁰ T. Chien, J.L. Hu, Renewable energy and macroeconomic efficiency of OECD and non-OECD economies, *Energy Policy* 35 (7) (2007) 3606 - 3615.
- ²¹ P. Sadorsky, Renewable energy consumption and income in emerging economies, *Energy Policy* 37 (10) (2009) 4021 - 4028.
- ²² Y. Chen, Factors influencing renewable energy consumption in China: an empirical analysis based on provincial panel data, *J. Clean. Prod.* 174 (2018) 605 - 615.

-
- ²³ J. Yang, W. Zhang, Z. Zhang, Impacts of urbanization on renewable energy consumption in China, *J. Clean. Prod.* 114 (2016) 443- 451.
- ²⁴ N. Apergis, J.E. Payne, Renewable energy consumption and growth in Eurasia, *Energy Econ.* 32 (6) (2010) 1392e - 397
- ²⁵ N. Apergis, J.E. Payne, Renewable energy consumption and economic growth: evidence from a panel of OECD countries, *Energy Policy* 38 (1) (2010) 656 - 660
- ²⁶ M.M. Rahman, E. Velayutham, Renewable and non-renewable energy consumption-economic growth nexus: New evidence from South Asia. *Renewable Energy* 147 (2020) 399 – 408.
- ²⁷ K.H. Nguyen, M. Kakinaka, Renewable energy consumption, carbon emissions, and development stages: some evidence from panel cointegration analysis, *Renew. Energy* 132 (2019) 1049 - 1057.
- ²⁸ Akhmat G, Zaman K, Shukui T. Balance of payment prospects and policy implications in light of energy sources in Pakistan. *Energy Educ Sci Technol Part B: Soc Educ Stud* 2013;5(4):655–86.
- ²⁹ World Bank, World Development Indicators, 2019.
- ³⁰ F. Hasanov, C. Bulut, E. Suleymanov, Review of energy-growth nexus: a panel analysis for ten Eurasian oil exporting countries, *Renew. Sustain. Energy Rev.* 73 (2017) 369 - 386.
- ³¹ P. Pedroni, Critical values for cointegration tests in heterogeneous panels with multiple regressors, *Oxf. Bull. Econ. Stat.* 61 (s 1) (1999) 653 - 670.
- ³² C. Kao, Spurious regression and residual-based tests for cointegration in panel data, *J. Econom.* 90 (1) (1999) 1 - 44.
- ³³ P. Pedroni, Panel cointegration: asymptotic and finite sample properties of pooled time series tests with an application to the PPP hypothesis, *Econom. Theor.* 20 (03) (2004) 597 - 625.
- ³⁴ R.F. Engle, C.W. Granger, Co-integration and error correction: representation, estimation, and testing, *Econometrica*, *J. Econom. Soc.* (1987) 251 - 276.
- ³⁵ A. Kasman, Y.S. Duman, CO2 emissions, economic growth, energy consumption, trade and urbanization in new EU member and candidate countries: a panel data analysis, *Econ. Modell.* 44 (2015) 97 - 103.
- ³⁶ M. Hamit-Haggar, Greenhouse gas emissions, energy consumption and economic growth: a panel cointegration analysis from Canadian industrial sector perspective, *Energy Econ.* 34 (1) (2012) 358 - 364.
- ³⁷ B. Ozcan, The nexus between carbon emissions, energy consumption and economic growth in Middle East countries: a panel data analysis, *Energy Policy* 62 (2013) 1138 - 1147.
- ³⁸ C. Kao, M.H. Chiang, On the estimation and inference of a cointegrated regression in panel data, in: B. Baltagi (Ed.), *Nonstationary Panels, Panel Cointegration, and Dynamic Panels (Advances in Econometrics)*, JAI Press, Amsterdam, 2000, pp. 161 - 178.
- ³⁹ W. Fan, Y. Hao, An empirical research on the relationship amongst renewable energy consumption, economic growth and foreign direct investment in China, *Renewable Energy* 146 (2020) 598 – 609.
- ⁴⁰ R. Mahadevan, J. Asafu-Adjaye, Energy consumption, economic growth and prices: a reassessment using panel VECM for developed and developing countries, *Energy Policy* 35 (4) (2007) 2481 - 2490.
- ⁴¹ Y. Hao, L.O. Wang, L. Zhu, M. Ye, The dynamic relationship between energy consumption, investment and economic growth in China's rural area: new evidence based on provincial panel data, *Energy* 154 (C) (2018) 374 - 382.
- ⁴² E.I. Dumitrescu, C. Hurlin, Testing for Granger non-causality in heterogeneous panels, *Econ. Modell.* 29 (4) (2012) 1450 - 1460.
- ⁴³ E. Dogan, F. Seker, Determinants of co2 emissions in the European Union: the role of renewable and non-renewable energy, *Renew. Energy* 94 (2016) 429 - 439.
- ⁴⁴ E. Koçak, A. Sarkgünesi, The renewable energy and economic growth nexus in black sea and Balkan countries, *Energy Policy* 100 (2017) 51 - 57.
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Appendix